

# Malhar Gudekar

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## EDUCATION

### University of Illinois Urbana Champaign

August 2024 - May 2026

Master of Science, Information Management

GPA: 3.8/4

• Coursework: Data Warehousing & Business Intelligence, Information Management, Information Consulting, Methods of Data Science

### University of Mumbai

June 2019 - May 2023

Bachelor of Engineering, Computer Science

GPA: 9.25/10

• Coursework: DBMS, Big Data Analysis, Data Warehousing and Mining, Project Management, NLP, Machine Learning, Social Media Analytics

## SKILLS

• **Analytical:** Data Analysis, Exploratory Data Analysis, Statistical Analysis, Trend Analysis, Root Cause Analysis, KPI Reporting, Data Validation, Data Cleaning, Anomaly Detection, ETL, Predictive Analytics

• **Programming / Querying:** Python, SQL, Pandas, NumPy, PySpark, PostgreSQL, Neo4j

• **Visualization / Reporting:** Power BI, Tableau, DAX, Excel, VBA, Dashboard Development, Data Visualization

• **Data / Cloud / Tools:** AWS S3, EC2, Redshift, EKS, FastAPI, Docker, Git, Kafka, pgVector, GCP

## WORK EXPERIENCE

### University of Illinois Urbana Champaign

May 2025 - Present

Research Assistant - iSchool

Champaign, USA

• Owned end-to-end design and development of **cross-platform system integrating** wearable data for 100+ users, enabling scalable **data ingestion** and analytics

• Designed and optimized **PostgreSQL architecture** for high-volume, production-scale datasets, improving query performance by 38%

• Built robust Python-based data analysis and validation pipeline, implementing back-testing logic and accuracy benchmarking across multi-source datasets, improving output accuracy by 43%

• Developed **backend workflows** for real-time, fault-tolerant processing of longitudinal health data with focus on reliability and scalability

Research Assistant - CHI AI in Infodemic Management

January 2025 - May 2025

• Designed and deployed highly **scalable NLP pipelines in Python** to process high-volume datasets, reducing processing latency by **41%**

• Built distributed data ingestion pipelines using **Kafka** and **Neo4j** Streams for real-time, high-throughput processing and pattern detection

• Developed production-grade **APIs** and deployment workflows to improve system reliability and reproducibility

• Enabled large-scale data processing across social media streams using **distributed system architecture** principles

### Business Intelligence Group

August 2025 - December 2025

Technical Consultant

Champaign, USA

• Built and deployed scalable, microservices-based (**RAG**) system using **FastAPI**, **PostgreSQL**, and vector embeddings

• Designed efficient, fault-tolerant **ETL** pipelines to process and index large document datasets for low-latency retrieval

• Improved system performance by **64%** through optimized indexing strategies and pipeline efficiency improvements

• Developed backend services enabling real-time, high-performance querying of unstructured data

### Swift Mobil Software Solutions Provider

July 2023 - October 2023

Data Analyst

Mumbai, IN

• Built interactive **Power BI** dashboards for **logistics** and **mobility KPIs**, reducing reporting time by **40%** and improving visibility into operational performance across transportation services

• Processed and optimized large-scale operational datasets using **Python** and **PySpark** to identify failure patterns, trends, and performance

• Developed SQL queries to perform data analysis, comparison, and validation across operational datasets, detecting quality issues and anomalies, speeding root-cause analysis by 26%

### Pscope Technologies Pvt. Ltd.

January 2023 - June 2023

Data Analyst

Mumbai, IN

• Developed and maintained **20+ Power BI** dashboards across Sales, Finance, and Operations for enterprise clients, applying **SQL**, **DAX**, and **statistical analysis** to surface KPI trends and support cross-functional business decision-making

• Automated and optimized complex Excel-based financial reporting workflows using VBA, implementing validation logic and scheduling, reducing report preparation time by 42% across weekly cycles

### Stu/Dio at Illinois

August 2025 - Present

Project Manager

Champaign, USA

• Led cross-functional team of **6** to deliver product updates for DeepCover, game, using user feedback and performance insights to guide data-driven feature decisions

• Tracked **sprint** progress, delivery metrics, and **QA issues** through **JIRA**-based workflows, improving cross-team visibility and supporting on-time milestone reviews

## PROJECTS

### Scalable ML System for Customer Risk Prediction

March 2025

[View Project](#)

• Performed exploratory data analysis and feature engineering on large-scale financial transaction datasets to identify behavioral patterns, spending trends, and risk signals for credit assessment

• Built and validated a multivariate regression and LightGBM model on time-series financial transaction data, performing back-testing and comparative evaluation to improve credit risk prediction precision by 20%

• Developed a risk-based decision framework incorporating model outputs and threshold validation, translating financial data analysis into actionable credit limit recommendations with measurable precision gains